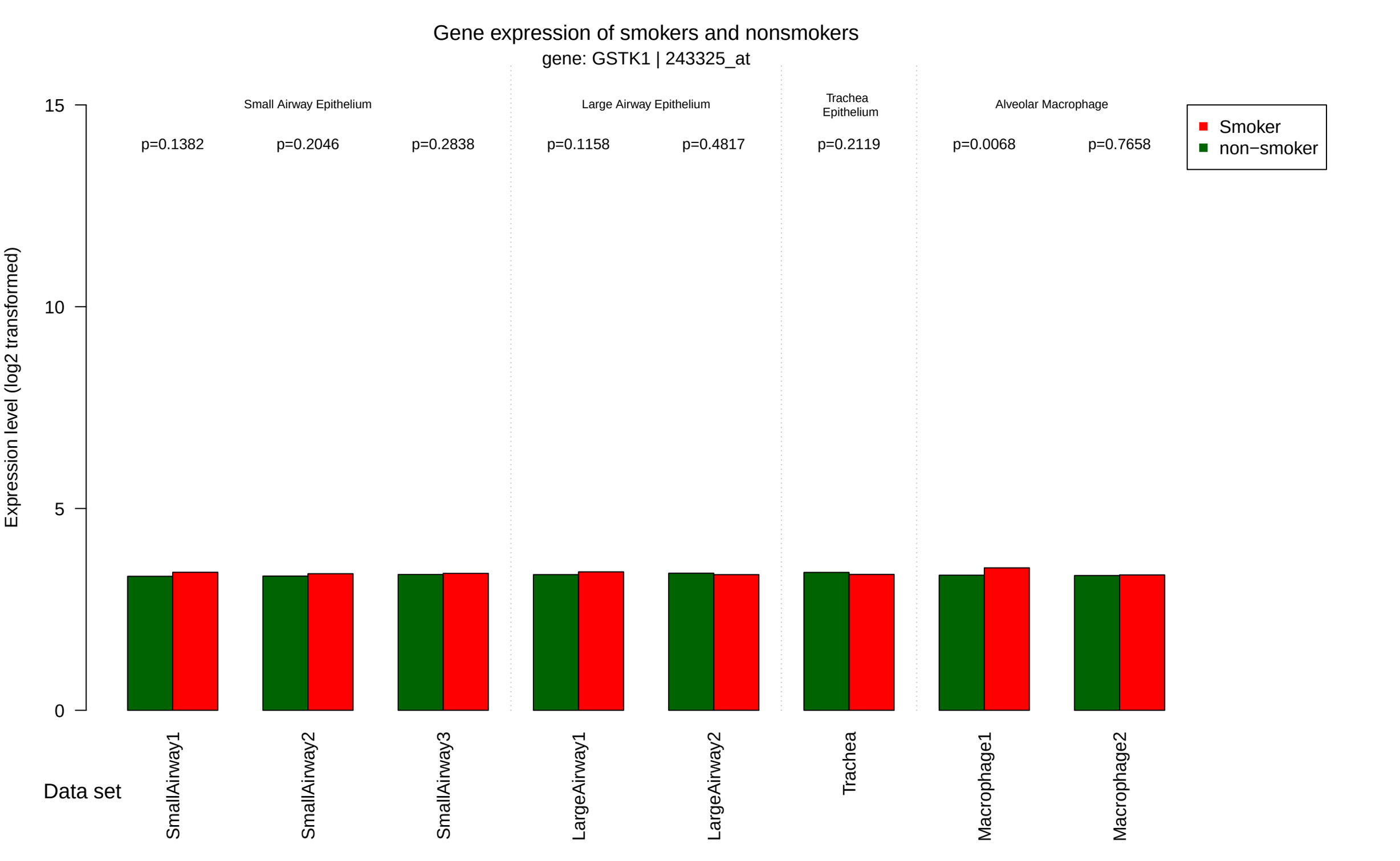
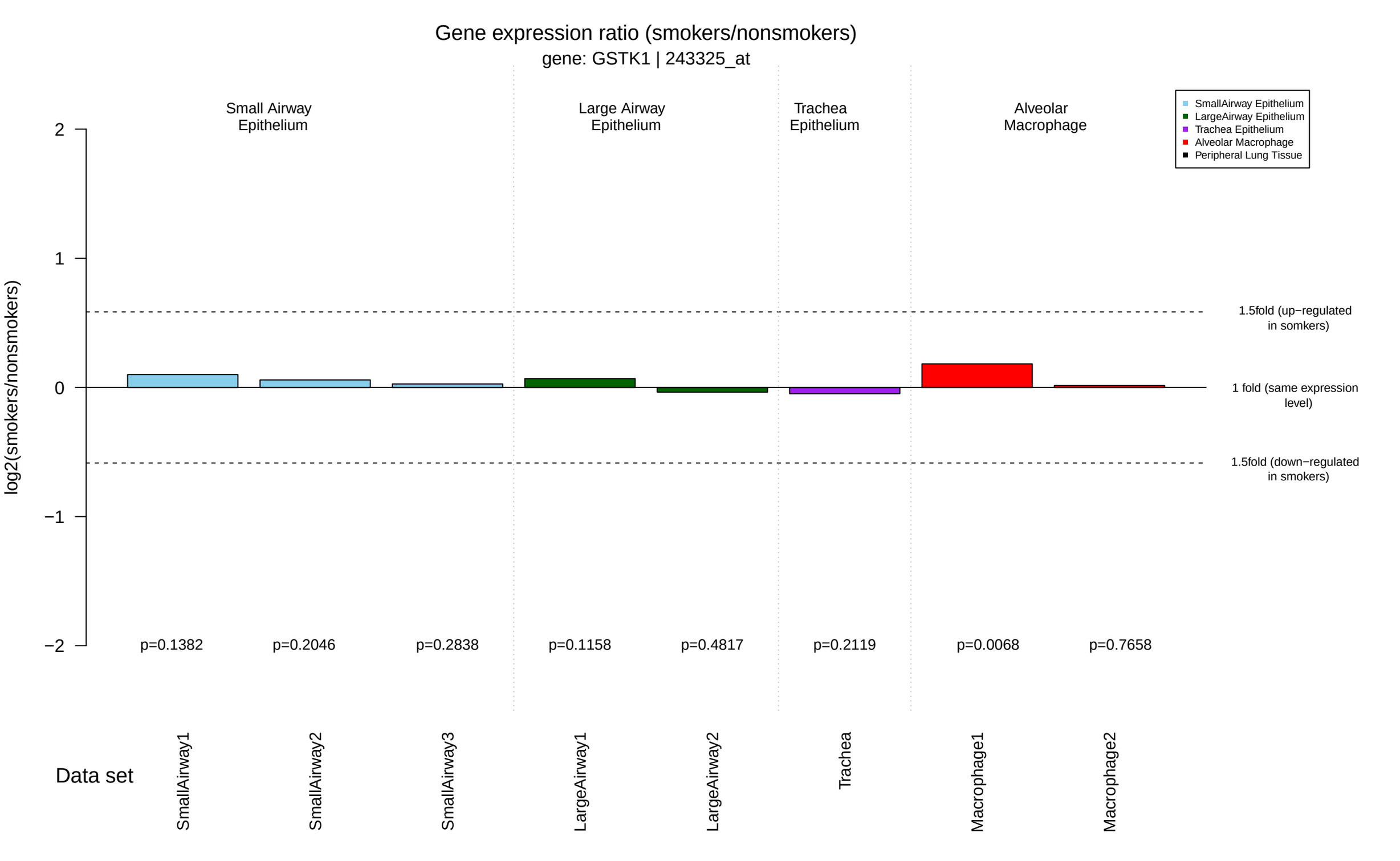
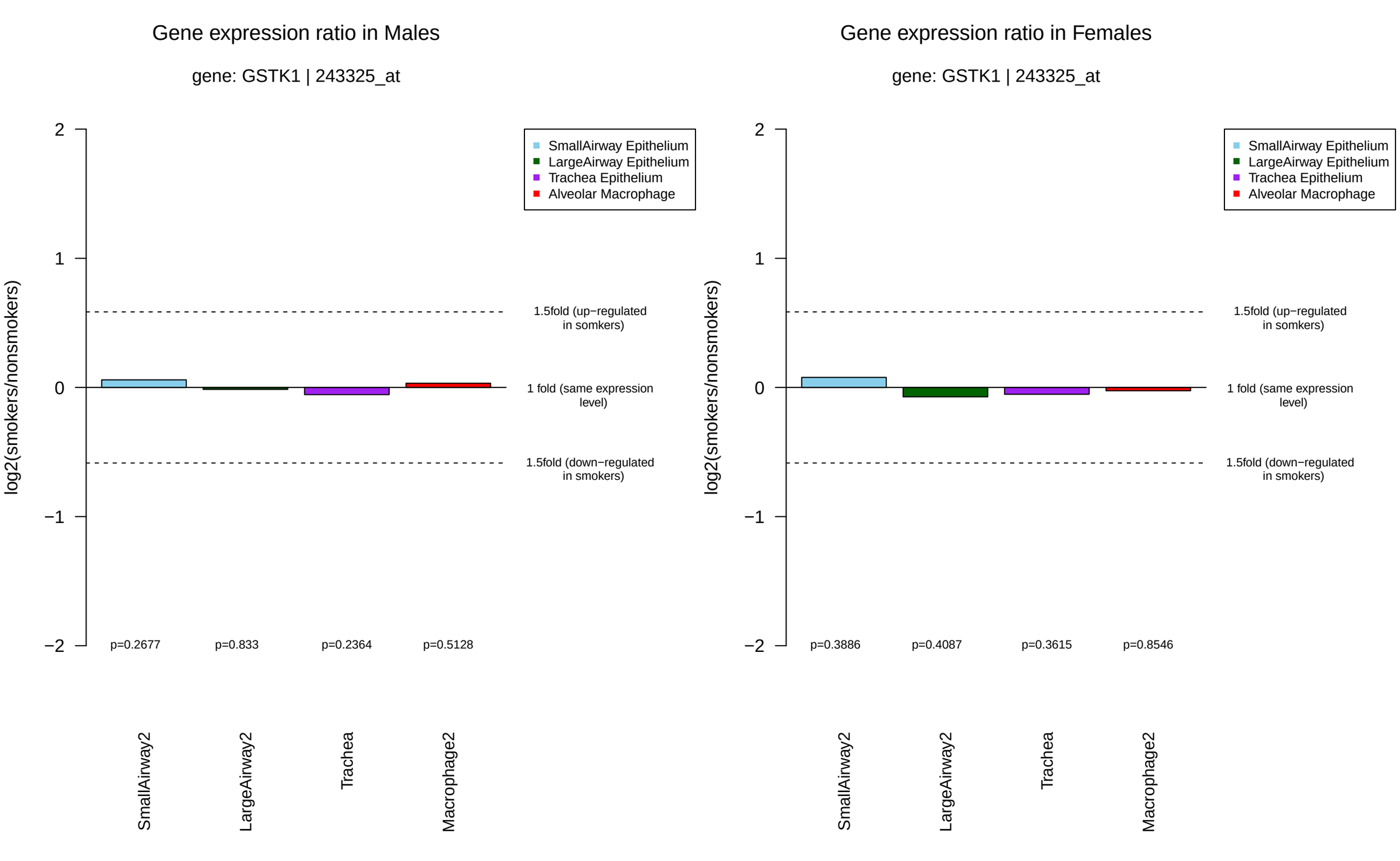
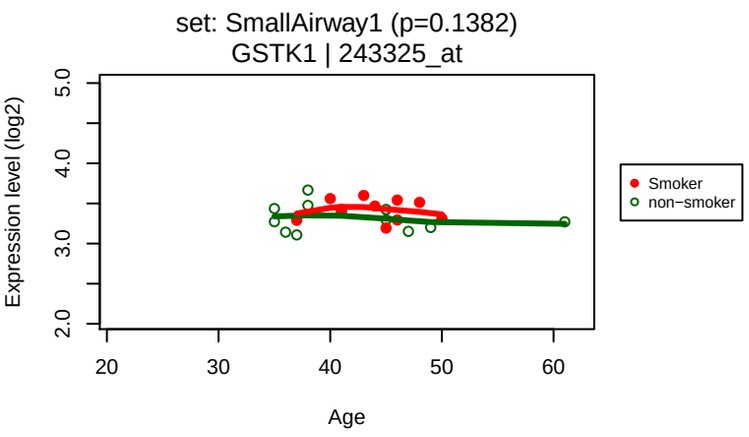
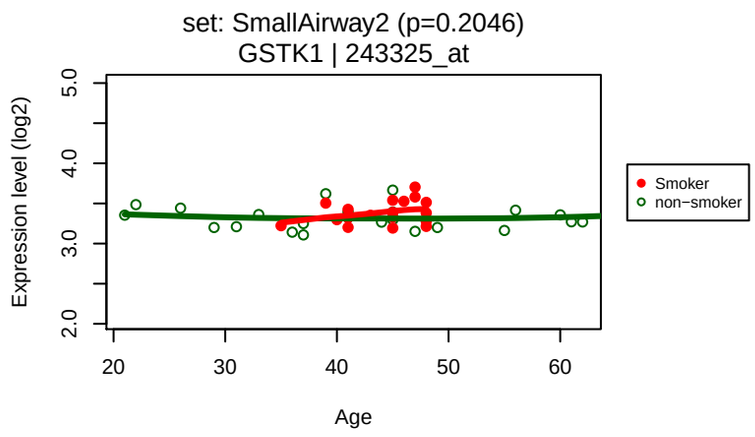
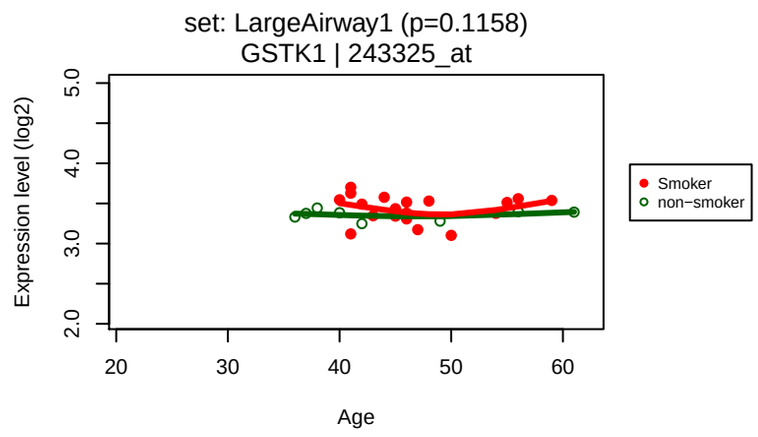

Gene expression of smokers and nonsmokers
gene: GSTK1 | 243325_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1382
p=0.2046
p=0.2838
p=0.1158
p=0.4817
p=0.2119
p=0.0068
p=0.7658
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

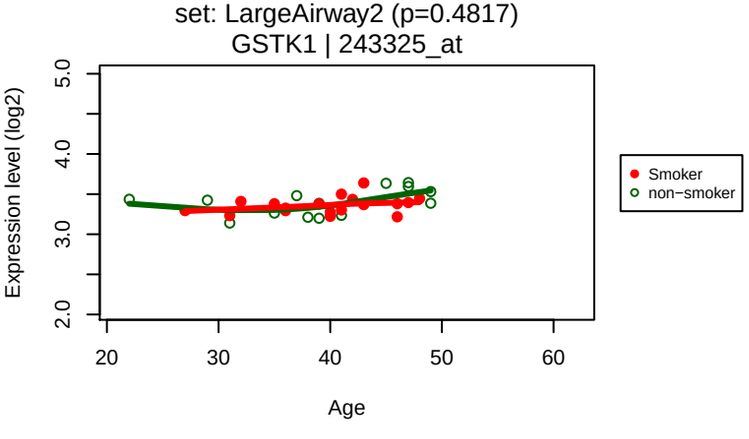

Gene expression ratio (smokers/nonsmokers)
gene: GSTK1 | 243325_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1382
p=0.2046
p=0.2838
p=0.1158
p=0.4817
p=0.2119
p=0.0068
p=0.7658
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

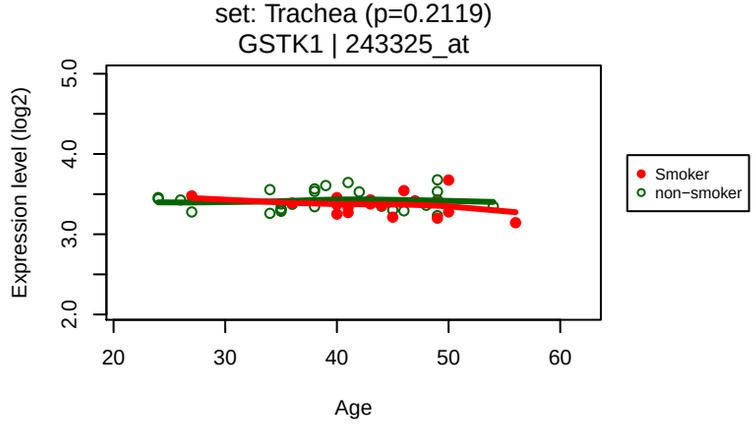

Gene expression ratio in Males
gene: GSTK1 | 243325_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2677
p=0.833
p=0.2364
p=0.5128
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSTK1 | 243325_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3886
p=0.4087
p=0.3615
p=0.8546
SmallAirway2
LargeAirway2
Trachea
Macrophage2

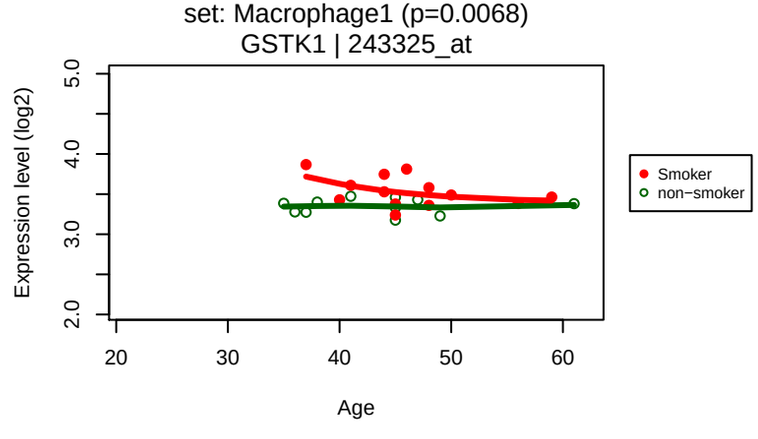

set: SmallAirway1 (p=0.1382)
GSTK1 | 243325_at
Expression level (log2)
Age
Smoker
non−smoker

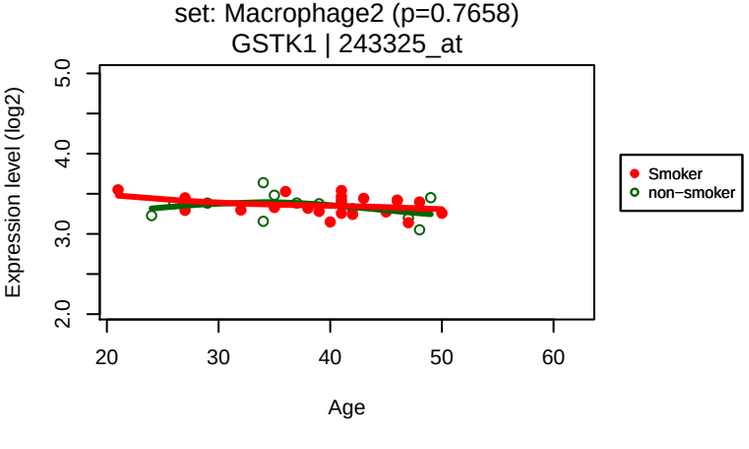

set: SmallAirway2 (p=0.2046)
GSTK1 | 243325_at
Expression level (log2)
Age
Smoker
non−smoker

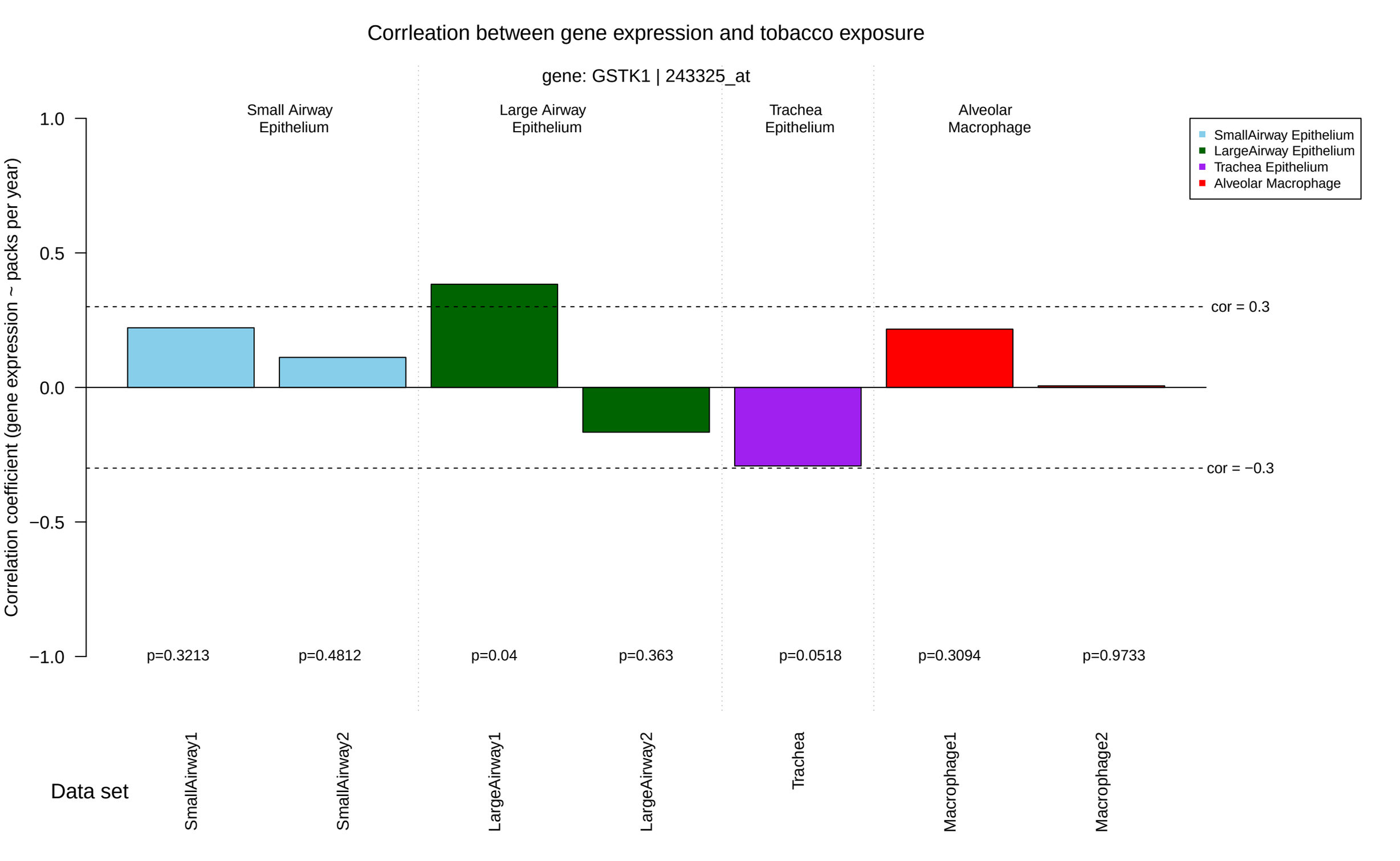

set: LargeAirway1 (p=0.1158)
GSTK1 | 243325_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4817)
GSTK1 | 243325_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2119)
GSTK1 | 243325_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0068)
GSTK1 | 243325_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7658)
GSTK1 | 243325_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSTK1 | 243325_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3213
p=0.4812
p=0.04
p=0.363
p=0.0518
p=0.3094
p=0.9733
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2